Multiview Calibrated Prototype Learning for Few-Shot Hyperspectral Image Classification

Chunyan Yu, *Member, IEEE*, Baoyu Gong, Meiping Song, *Member, IEEE*, Enyu Zhao, and Chein-I Chang, *Life Fellow, IEEE*

***Abstract*— Despite continuing to progress in hyperspectral image classification (HSIC) based on deep learning, the classification accuracy is limited to further improve in the absence of labeled samples. To address this issue, the metric-based prototypical networks for few-shot learning have enjoyed widespread popularity. However, the conventional prototypical networks are vulnerable to the selected examples and fail to accomplish representative predictions for the prototypes in complicated situations. In this article, we propose a multiview calibrated prototype-learning framework for few-shot HSIC, which consists of three rectified strategies from different views to improve the robustness of prototypes in the embedding space. Specifically, the calibrated aggregation network is the first presented to calibrate the representations with local patches' aggregation for the enhancement of the prototypes. Moreover, to improve the compactness of the intraclass expression, the calibrated metric learning with regularization terms is designed to strengthen the discrimination of the prototypes. Furthermore, we calibrate the feature distribution of supervised samples by transferring statistical knowledge to eliminate the local bias in the test phase. The extensive experimental results and analysis of three hyperspectral image datasets demonstrate the superiority of the proposed architecture compared with other advanced methods.**

***Index Terms*— Feature distribution, feature extraction, few-shot hyperspectral image classification (HSIC), metric learning, prototype learning.**

I. Introduction

HYPERSPECTRAL remote sensing technology collects the union of imagery and spectrum containing abundant spectral bands. With the distinguishable ability of spectra varied from different ground objects, hyperspectral image (HSI) is proven to be of great significance to the areas related to precision agriculture, military reconnaissance, and geological exploration, in which HSI classification (HSIC) is an active and challenging task due to dimensionality curse and complicated scenario.

Traditional machine learning methods, such as supporting vector machine (SVM) [1], principal component analysis (PCA) [2], and active learning (AL) [3], have greatly promoted the progress of HSIC to a certain extent. More specifically, iterative support vector machine (IVSM) [4] was proposed to combine the spatial and spectral information and correct the training examples for the performance of HSIC. The HSIC accuracies obtained using extended morphological profiles (MPs) built from the features of nonlinear PCA (NLPCA) were investigated in [5]. Nevertheless, achieving high-performance classification is still a critical problem due to the imbalance between the high dimensionality and the shallow structure of the traditional models. Recently, owing to the in-depth exploration and application of deep learning (DL), significant improvements have been witnessed in HSIC [6], [7], [8], [9], [10], [11], [12]. In the early phase, the DL models focused on the spectrum of HSI for feature extraction and network training. In particular, a deep convolutional neural network (CNN) is employed for HSIC directly in the spectral domain in [13]. Jia et al. [14] reconstructed the spectral feature image by extracting features in different layers from convolution layers and pool layers. Along with the deep-going of research, more attention is drawn to the combination of spatial–spectral information to accomplish the classification tasks, which provides an effective idea for HSIC [15], [16], [17], [18], [19], [20], [21], [22], [23]. Typically, Sun et al. [20] proposed a powerful detector that adaptively learns the optimal separation between the target and the background, which is completely free from the constraints of specific model assumptions. The proposed method could fully excavate and precisely quantify the spectral differences of various land covers to accurately locate the target of interest. Zhang et al. [21] proposed group sparse coding for each pixel and its spatial neighbors to obtain sparse representation vectors for the SVM classification of HSI. 3DOC-SSAN was proposed in [22] to deeply exploit spatial–spectral information for HSIC based on a 3-D octave convolution model and two attention models. Besides, Kanthi et al. [23] divided the HSI data into 3-D patches and fed them into a 3-D-deep CNN model for utilizing both spectral and spatial information. Nevertheless, most of the existing convolutional network models depend on manually labeled sample

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data, whereas collecting labeled samples is labor-intensive and inefficient for HSIC. Consequently, seeking out an effective solution to HSIC with a shortage of labeled samples has become a research hotspot.

As a remedy, a growing number of studies have been explored to find alternative ways to tradeoff between the lack of labels and the complexity of the models. Few-shot learning inspired by the ability of humans to learn quickly is a popular solution in this regard with a typical characteristic to recognize new classes when limited labeled samples are available. Especially, model-based few-shot learning, optimization-based few-shot learning, and metric-based few-shot learning constitute three principal types of the few-shot learning field. The model-based method focuses on the structural design of the model to update the parameters on a few samples. Typically, MAML [24] and TAML [25] only learn initial models and attempt to produce good generalization performance on new tasks. Adjusting the optimization strategy to complete the classification is the core strategy of optimization-based few-shot learning. In particular, the LSTM-based metalearner was proposed in [26] to learn the exact optimization algorithm. Elsken et al. [27] proposed a META-learning based Neural Architecture Search (METANAS) approach to optimize a meta-architecture and the metaweights during metatraining. Significantly, metric-based few-shot learning is a kind of widely prevalent settlement that employs distance metrics and nonparametric classifiers, such as the nearest neighbor (NN) classifier to accomplish classification tasks through the measured similarity. In this kind of scenario, the Siamese network [28], the matching network [29], and the prototypical network [30] have all been proposed for a few-shot classification with stable performance. Nowadays, many achievements related to HSIC [31], [32], [33], [34], [35] in few-shot learning occupy a significant position relying on unremitting efforts. For instance, Ma et al. [36] proposed an HSIC method based on a task-specific learning network. The AwGCN model was built in [37] that aimed to explore the internal relationship between data for semisupervised label propagation. Wang et al. [38] and Huang and Chen [39] both employ the Siamese network to solve the problem of limited labeled samples for HSIC.

Among the metric-based mechanism, prototype learning is an emblematical pattern of few-shot classification due to the simplicity and practicality. In general, prototype learning permits each class to possess a concise representation irrelevant to the number of samples and avoids storing the whole support set for predictions. The goal of the prototypical network is to learn a metric space in which classification tasks can be achieved by calculating the distance to each prototype. Particularly, the Euclidean distance as a Bregman divergence [30] outperforms the commonly used cosine similarity in distance metrics. Nowadays, many scholars have expressed great interest in prototype learning for HSIC and put forward a series of excellent algorithms [40], [41], [42]. Notably, Liu et al. [43] proposed a DFSL method for HSIC to train a network by episodes, and the testing samples are classified by the NN classifier. In [44], a global representation learning strategy was exploited to train the network to transfer samples from the original data space to the embedding space, and a vector of global prototype representation was learned for each class. In order to classify the mixed pixels existing in HSI, SSPN in [45] solved the problem of lacking labeled samples for HSI through a novel prototype representation in metric space. Xi et al. [46] proposed a deep prototypical network with hybrid residual attention, and a novel loss function was utilized to increase the intraclass compactness.

Nevertheless, the previous implementation ignores the deviations caused by directly taking average prototypes from only a handful of labeled data in the embedding space. Moreover, the average vectors hinder the model from learning the more appropriate representations of the randomness of sample selection. In this article, a novel multiview calibrated prototypical network (MCPN) for HSIC with rectification tactics from multiple perspectives is proposed to promote the effectiveness of prototypes. First, the calibrated acquisition of prototypes is accomplished through the aggregation of HSI patches to achieve persuasive representations. In addition, the application of calibrated metric learning (CML) is motivated to strengthen regularization and increase the intraclass compactness and the interclass discrepancy. Furthermore, to relieve the distribution difference between the supervised samples and global data, we calibrate the feature distribution with transferred statistics in the test phase.

In summary, the main contributions of this article are listed as follows.

1) We present a novel multiview calibrated prototype-learning framework for few-shot HSIC, which is accomplished with an innovative calibrated representation method to compute prototypes. Notably, the calibrated aggregation network (CAN) is the first attempt to convert the acquisition of prototypes from the average vector into local HSI patch aggregation.
2) We design a calibrated metric strategy consisting of prediction loss, discriminative loss, and contrastive loss to strengthen the intraclass similarity and interclass discrepancy simultaneously for promoting the regularization and the generalization ability of the prototype learning.
3) We calibrate the feature distribution by correlating the concentration and dispersion with semantic similarity in the test phase, which tackles the local deviation issue by aligning the characteristic distribution of supervised samples and overall data.

The rest of this article is organized as follows. In Section II, the proposed approach in this article is introduced in detail. Section III describes the experimental results and analyses the experimental results. Section IV draws the conclusion.

II. Proposed Approach

As exhibited in Fig. 1, the proposed framework of the multiview calibrated prototype-learning model mainly consists of three major parts: the initial prototype computation (IPC) with feature extraction network (FEN), the calibrated prototype representation (CPR) with the CAN, and the CML. Specifically, the spectral channels of the datasets are first reduced to 20 principal bands from PCA to maintain maximum spectral–spatial information. Afterward, the initial prototypes

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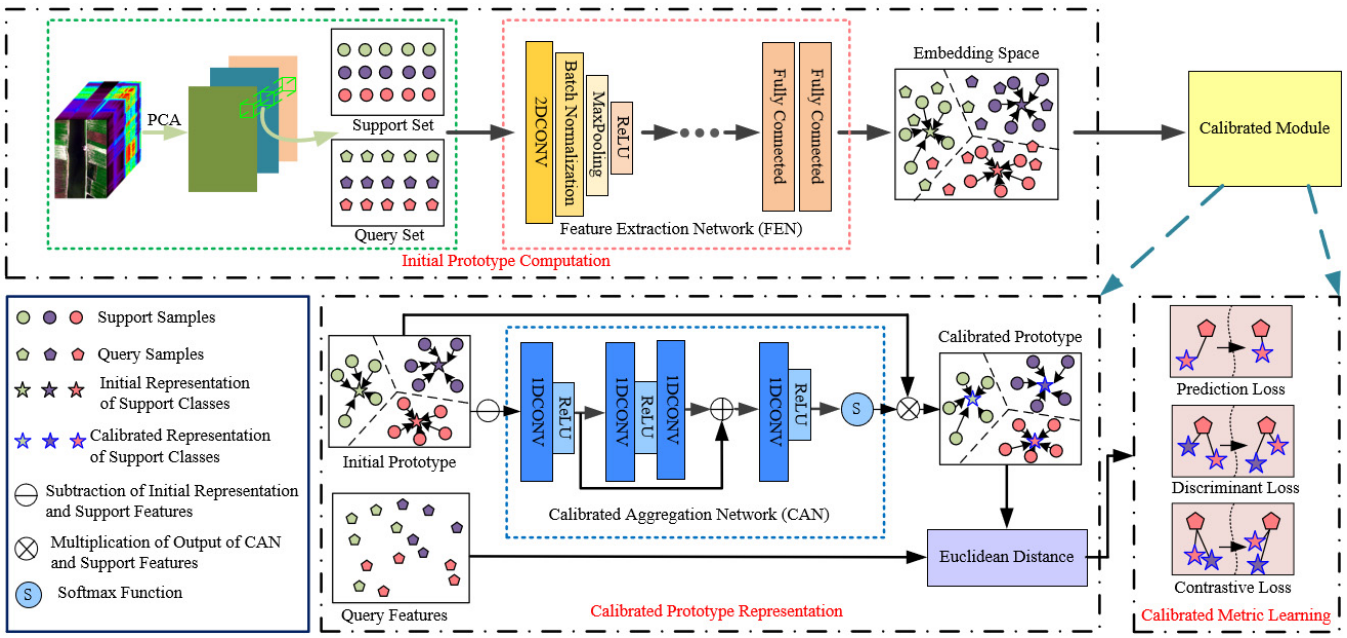


Fig. 1. Architecture of the proposed calibrated model in this article. The complete training process is exhibited for an episode comprised of $N_c = 3$, $N_s = 5$, and $N_q = 5$. The overview of the proposed model consists of three major components: the initial representation computation for extracting features with FEN and calculating the initial prototypes, the CPR to calibrate the prototypes with CAN, and the CML aiming to strengthen the intraclass similarity and interclass discrepancy, and promote the discrimination of the learned prototypes.

are calculated by the embedding features extracted through FEN. Next, this article adopts the calibrated module including the CPR and the CML to improve the prediction precision and robustness of the prototypes. In particular, to complete the prototype calibration, the differences between the initial prototype and the support features are fed to the CAN to obtain the aggregation parameters. In addition, the CML with three regularization terms is employed to further increase the intraclass compactness and interclass discrepancy.

A. Initial Prototype Computation

In this article, the embedding space suitable for few-shot HSIC is learned by the episode strategy mimicking the few-shot task. Especially, the mapping function of the embedding space parameterized by the deep network FEN parameterization is expressed as $f_{\theta_F}: \mathbb{R}^D \rightarrow \mathbb{R}^M$, where θ_F is a learnable parameter of the embedding space, D denotes the spectral dimension of the original samples, and M refers to the dimension of embedding features.

The implementation of the feature extractor is based on 2-D CNN. Precisely, the network consists of three convolution blocks, and each block is composed of one convolution layer, a batch normalization layer, one max-pooling layer (not part of the third block), and an Rectified Linear Unit (ReLU) activation layer. Eventually, the optimized features are fed into two fully connected layers for feature compaction.

In the few-shot classification task, a part of the samples is randomly chosen to serve as the support set, and another subset is acted as the query set. Specifically, S_k and Q_k are assumed to represent the support set and the query set of the kth class in this article, respectively. S_k is expressed as follows:

$$S_k = \{(x_1, y_1), \ldots, (x_i, y_i), \ldots, (x_{Ns}, y_{Ns})\} \tag{1}$$

where x_i is a D-dimensional vector in the original data space, y_i is the corresponding label, and N_s is the number of the labeled samples of the kth class contained in S_k. Similarly, $Q_k = \{(x, y)\}^{Nq}$, where N_q denotes the number of samples in Q_k.

Accordingly, the initial prototype related to the kth class is calculated by averaging the embedding features, and the computation function is defined as follows:

$$C_k = \frac{1}{|S_k|} \sum_{(x_i, y_i) \in S_k} f_{\theta_F}(x_i) \tag{2}$$

where $f_{\theta_F}(\cdot)$ is the embedding function that is responsible for the output of the sample x_i.

B. Calibrated Prototype Representation With CAN

Due to the randomness of the selection strategy and the significance of prototype representation, the initial prototypes obtained by the average vector are less persuasive. Furthermore, to permit the calibrated prototypes to cover and represent the whole data for each class, the calibrated aggregation method is implemented by a convolution network with a residual block. In our model, CAN is designed to assist the implementation of calibrated representations with the essence of utilizing the difference vector between the initial representation and support features to construct the basis for aggregation. Notably, since 1-D embedding features are acquired after the FEN module, the differences between embedding projection and the initial prototypes are input into CAN with 1-D convolution for calibration. Specifically, CAN is composed of four layers of 1-D convolution, and each layer extracts the representation of samples in high-dimensional space through a 1×1 convolution kernel. In addition, the

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design of the residual structure avoids the disappearance of the gradient and accelerates the convergence process of the network.

The aggregation parameter for the sample x_i is calculated as follows:

$$w_i = g_{\theta_C}(C_k - f_{\theta_F}(x_i)) \tag{3}$$

where $g_{\theta_C}(\cdot)$ represents the feature space with the parameter θ_C of CAN, which provides the basis for HSI patch aggregation.

Subsequently, the prototype of the kth class is calibrated by the definition as follows:

$$C'_k = \sum_{i \in Ns}^{Ns} f_{\theta_F}(x_i) w_i. \tag{4}$$

C. Calibrated Metric Learning

In this article, we propose CML with regularization terms to strengthen the discrimination of the learned prototypes. Specifically, minimizing the loss function through the Adam optimizer is to stimulate the network to generate discriminative embedding space with increased interclass distance and intraclass compactness. The loss function is described in detail.

Prediction Loss: To measure the similarity between query samples and prototypes, we adopt the prediction loss function based on the Euclidean distance. Since the Euclidean distance is inversely proportional to the similarity, the intraclass similarity is strengthened by minimizing the prediction loss function. The probability of the query sample x belonging to the kth class is represented as follows:

$$p(y = k|x) = \frac{\exp\left(-d\left(f_{\theta_F}(x), C'_k\right)\right)}{\sum_{k=1}^{N_C} \exp\left(-d\left(f_{\theta_F}(x), C'_k\right)\right)} \tag{5}$$

where $d(\cdot)$ is the Euclidean distance function and $\exp(\cdot)$ denotes the exponential function.

The prediction loss function is defined as the negative log-probability

$$J = -\log(p(y = k|x)). \tag{6}$$

Discriminant Loss: To satisfy a more compact expression and further separate the query features from other categories simultaneously, a discriminant loss is proposed in our paper that is defined as follows:

$$J_{\mathrm{DL}} = \frac{d\left(f_{\theta_F}(x), C'_k\right)}{d\left(f_{\theta_F}(x), C'_l\right)} \tag{7}$$

where C'_k is the corresponding prototype of the query sample and C'_l is the negative prototype of the category closest to x. In the learning process, $d(f_{\theta_F}(x), C'_k)$ is decreased, and $d(f_{\theta_F}(x), C'_l)$ is increased when the discriminant loss function is minimized.

Contrastive Loss: Both intraclass similarity and interclass discrepancy are crucial for metric learning in few-shot HSIC. To further accomplish a more clustered embedding space, we apply the idea of contrastive learning with the integration of triplet loss to the calibration of the metric criterion. On the one side, we compute the intraclass variation by consideration of the query set and the related prototype specifically. On the other side, we consider that similar classes generate more contribution to the interclass discrepancy. Thus, we depended on the nearest category to calculate the interclass difference. In the implementation, all the measures are calculated by the Euclidean distance. Especially, contrastive knowledge is generated by measuring the similarities between the query set and the prototypes. In that case, intraclass similarity and interclass discrepancy are ensured by learning the samples with little difference. The contrastive loss based on triplet loss is expressed as follows:

$$J_C = \begin{cases} 0, & \text{if } d\left(f_{\theta_F}(x), C'_l\right) - d\left(f_{\theta_F}(x), C'_k\right) > \alpha \\ d\left(f_{\theta_F}(x), C'_k\right) - d\left(f_{\theta_F}(x), C'_l\right) & \\ \quad + \alpha, \quad \text{else} & \end{cases} \tag{8}$$

where α is to define the minimum margin between two Euclidean distances. When J_C is 0, the distance between $f_{\theta_F}(x)$ and C'_k is far less than that between $f_{\theta_F}(x)$ and C'_l without the contrastive optimization. When J_C is greater than 0, the distance between $f_{\theta_F}(x)$ and C'_l is smaller, which requires a minimized loss value to reduce the misclassification rate.

Correspondingly, the total loss of the proposed framework is formalized as

$$L = J + \gamma J_{\mathrm{DL}} + \eta J_C \tag{9}$$

where γ and η are the regularization coefficients affecting the weight of discriminant loss and contrastive loss.

In summary, the steps of the training episode with prototype calibration are outlined as in Algorithm 1, in which N_c denotes the number of the selected classes for a training episode, S_k and Q_k express the support set and the query set of the kth class, respectively, and N_s denotes the number of labeled samples in S_k.

D. Calibrated Distribution

Furthermore, as displayed in Fig. 2, the calibrated distribution (CD) is employed for the supervised samples in the testing phase to avoid the biased distribution caused by the few-shot. In the implementation, we attempt to modify the prototypes with the distribution information computed in terms of prior knowledge. Notably, all samples are employed to participate in the calculation of the prior distribution. In detail, we assume that the feature distribution of all categories is Gaussian distribution [47] in the embedding space. First, we judge the central tendency and dispersion degree of each category. Next, the statistical knowledge is transferred from the whole class to the supervised samples. Finally, we calibrate and standardize the samples with the transferred information to align the characteristic distribution.

Specifically, the mean of the feature vector related to the ith class is computed from the mean of each dimension in the vector. The calculation formula is given as follows:

$$\mu_i = \frac{\sum_{j=1}^{n_i} f_{\theta_F}(x_j)}{n_i} \tag{10}$$

where n_i is the number of samples of the ith class.

Covariance is utilized to indicate the extent to which different eigenvectors change in the same or reverse direction.

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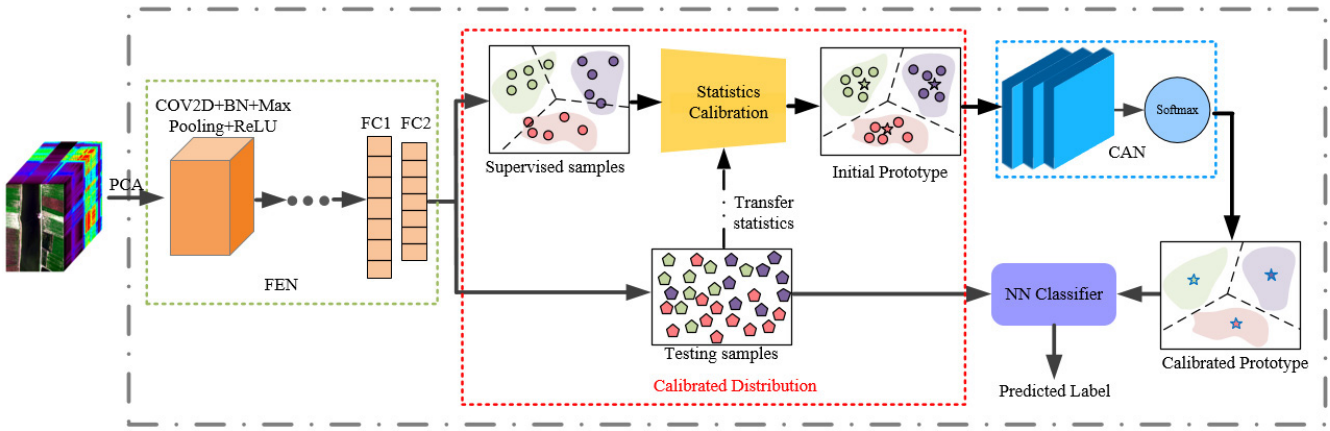


Fig. 2. Procedural diagram of classification task for the testing data. The CD module with five supervised samples per class and the labels predicted by the NN classifier.

Algorithm 1 Training Episode Algorithm

Input: The support set S_k and the query set Q_k.
Output: The parameters θ_F and θ_C.
Randomly Initialize: θ_F, θ_C.
Randomly select Nc classes and Ns support samples per class to construct the training episode.
For k in $\{1, \ldots, N_c\}$ **do**
Feed S_k and Q_k into FEN to extract the embedding features
$C_k \leftarrow$ Compute the mean of support embedding features as the initial prototype using Equation (2)
For i in $\{1, \ldots, N_s\}$ **do**
$w_i \leftarrow$ Calculate the aggregation parameter by CAN as Equation (3)
End
$C_k' \leftarrow$ Compute the calibrated prototype representation using Equation (4)
End
$L \leftarrow 0$ Initialize loss
For k in $\{1, \ldots, N_c\}$ **do**
For (x, y) in Q_k **do**
$J \leftarrow$ Calculate the prediction loss as Equation (6)
$J_{\mathrm{DL}} \leftarrow$ Calculate the discriminative loss as Equation (7)
$J_C \leftarrow$ Calculate the contrastive loss as Equation (8)
$L \leftarrow L + (J + \gamma J_{\mathrm{DL}} + \eta J_C)$ Update the total loss
End
End
$\theta_F, \theta_C =\leftarrow -\nabla_{\theta_F, \theta_C}(L)$ Update θ_F, θ_C with Adam optimizer

The covariance matrix of the ith class is calculated by the following formula:

$$\Sigma_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (f_{\theta_F}(x_j) - \mu_i)(f_{\theta_F}(x_j) - \mu_i)^T. \quad (11)$$

The statistics calibration of the supervised samples is expressed as follows:

$$\mu' = \frac{\mu_i + f_{\theta_F}(x)}{2} \quad (12)$$

$$\sum{}' = \sum_i + \varepsilon \quad (13)$$

where x is the supervised sample in the ith class and ε is a hyperparameter that determines the dispersion degree of features from the CD. Particularly, since the computation needs the label of the category, the calculations are regarded as the preprocessing operation, which is unrelated to the prediction of the test samples practically.

The feature vectors of the supervised samples after calibration and standardization are calculated as follows:

$$f_{\theta_F}(x)' = \left(\frac{f_{\theta_F}(x) - \mu'}{\sum{}'} \right) \sum_i + \mu_i. \quad (14)$$

Afterward, the final prototypes are computed by (4) and (14), which are representative enough to efficiently accomplish the classification task.

E. Classification of HSI Based on MCPN

In this section, we introduce the few-shot HSIC procedure based on MCPN in detail. To be specific, all samples are first fed into the pretrained feature extractor in the classification task of the testing datasets. Next, a subset of samples in each class is chosen to carry the role of supervised samples, and the feature distribution is calibrated with the transferred statistics. Furthermore, the final representations are derived from the calibrated features processed by local patches' aggregation. Eventually, the testing samples are classified by the NN classifier according to the distance from the calibrated prototypes.

In summary, the pseudocode of the proposed MCPN for few-shot HSIC is listed in Algorithm 2, where N denotes the number of classes in the testing dataset.

III. Experiments and Analysis

A. Data Description

The first dataset is the WHU-Hi-LongKou, which is acquired by the RSIDEA Research Group, Wuhan University, Longkou, Hubei, China, in 2018. The false-color image and the ground-truth image containing nine classes and 550 × 400 pixels are shown in Fig. 3(a) and (b). Each pixel has 270 bands of coverage ranging from 400 to 1000 nm.

The second one is the Houston dataset that is provided by the Hyperspectral Image Analysis Group and the Airborne Laser Mapping Center (NCALM) funded by the National

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Algorithm 2 MCPN for Few-Shot HSIC

Input: The Testing set
Output: The Predicted Label
Randomly select L samples as Supervised Samples
For i in $\{1, \ldots, N\}$ **do**
$\mu_i \leftarrow$ Compute the mean of the feature vector in the ith class as Equation (10)
$\Sigma_i \leftarrow$ Compute the covariance of the ith class according to Equation (11)
For t in $\{1, \ldots, \mathrm{L}\}$ **do**
$\mu' \leftarrow$ Compute the average value of the supervised sample using Equation (12)
$\sum' \leftarrow$ Compute the covariance of the supervised sample using Equation (13)
$f_{\theta_F}(x)' \leftarrow$ Compute the calibrated feature of the supervised sample as Equation (14)
End
Calculate the mean of calibrated features as the initial prototype
Calculate the calibrated prototype representation through the pre-trained CAN by the method mentioned in Algorithm 1
End
Predicted Label $\leftarrow$ Classify the testing samples by NN Classifier according to the distance to each calibrated prototype

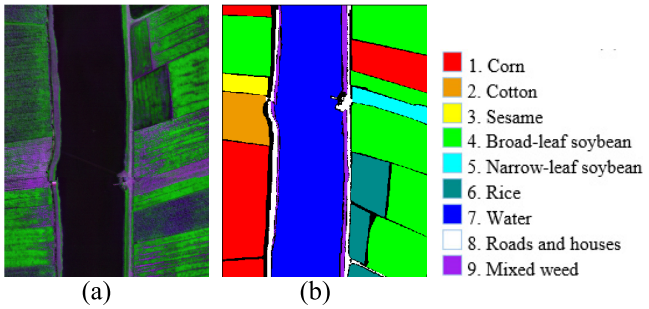


Fig. 3. Data of WHU-Hi-LongKou. (a) False-color image (bands 46, 235, and 118). (b) Ground-truth image.

Science Foundation (NSF) of the University of Houston. The dataset was originally used for the scientific purpose of the 2013 IEEE GRSS Data Fusion Competition. The size of the data is 349 × 1905 with 144 bands covering 15 classes as illustrated in the false-color image and the ground-truth image in Fig. 4.

The Botswana dataset is derived from a series of data obtained by NASA's EO-1 satellite in Botswana from 2001 to 2004. The sensor on EO-1 obtained 30-m pixel resolution data in 242 wavebands, covering a waveband of 400–2500 nm with 10-nm intervals. After removing the noise bands, the remaining 145 bands are used as candidate features. The false-color image and the ground-truth image cover 14 types of ground objects are shown in Fig. 5(a) and (b), respectively.

The details of the three datasets are illustrated in Table I.

B. Experimental Configuration

The implementation of all the models has been accomplished in the platform of PyTorch 1.2 with Python 3.7 on

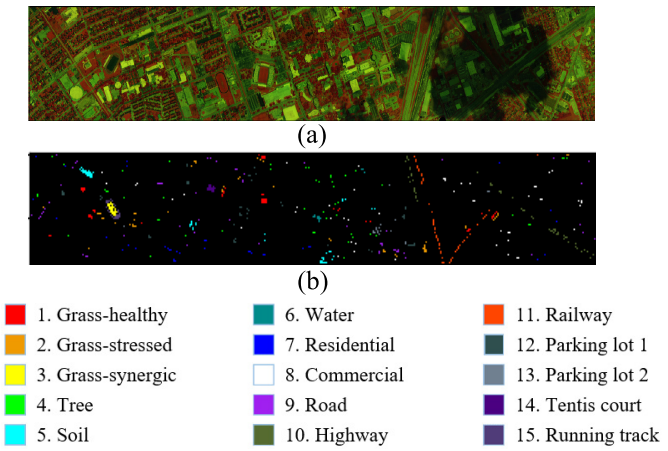


Fig. 4. Data of Houston. (a) False-color image (bands 78, 25, and 129). (b) Ground-truth image.

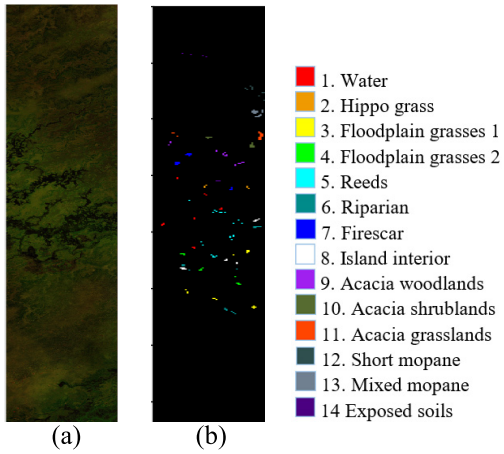


Fig. 5. Data of Botswana. (a) False-color image (bands 26, 73, and 88). (b) Ground-truth image.

TABLE I
DETAILS OF THREE DATASETS

Dataset	WHU-Hi-LongKou	Houston	Botswana
Spatial Size	550×400	349×1905	1476×256
Num. of Bands	270	144	145
Num. of Classes	9	15	14
Wavelength (nm)	400-1000	364-1046	400-2500
Areas	LongKou, China	Houston, USA	Botswana

a machine equipped with an Intel I5-8400 CPU, an Nvidia GTX 1050 Ti GPU, and 8-GB RAM. Due to the spatial correlation, interspectral correlation, and band correlation between adjacent bands, there is much redundant information. Therefore, the data should be processed by PCA before training, which can not only preserve the original data information but also improve the efficiency of network training. In the following experiment, the number of bands after PCA is fixed to 20, and the image size of the training samples is 9 × 9.

The WHU-Hi-LongKou dataset adopts the random sampling strategy to select 1% labeled samples as the training set, and as for the Houston dataset and the Botswana dataset, 10% of samples are selected. Next, a class subset is randomly selected from the training set to form an episode to calculate gradients and update parameters. Specifically, several samples are randomly chosen to serve as the support set and a subset of the rest acted as the query set. In this article, the number of

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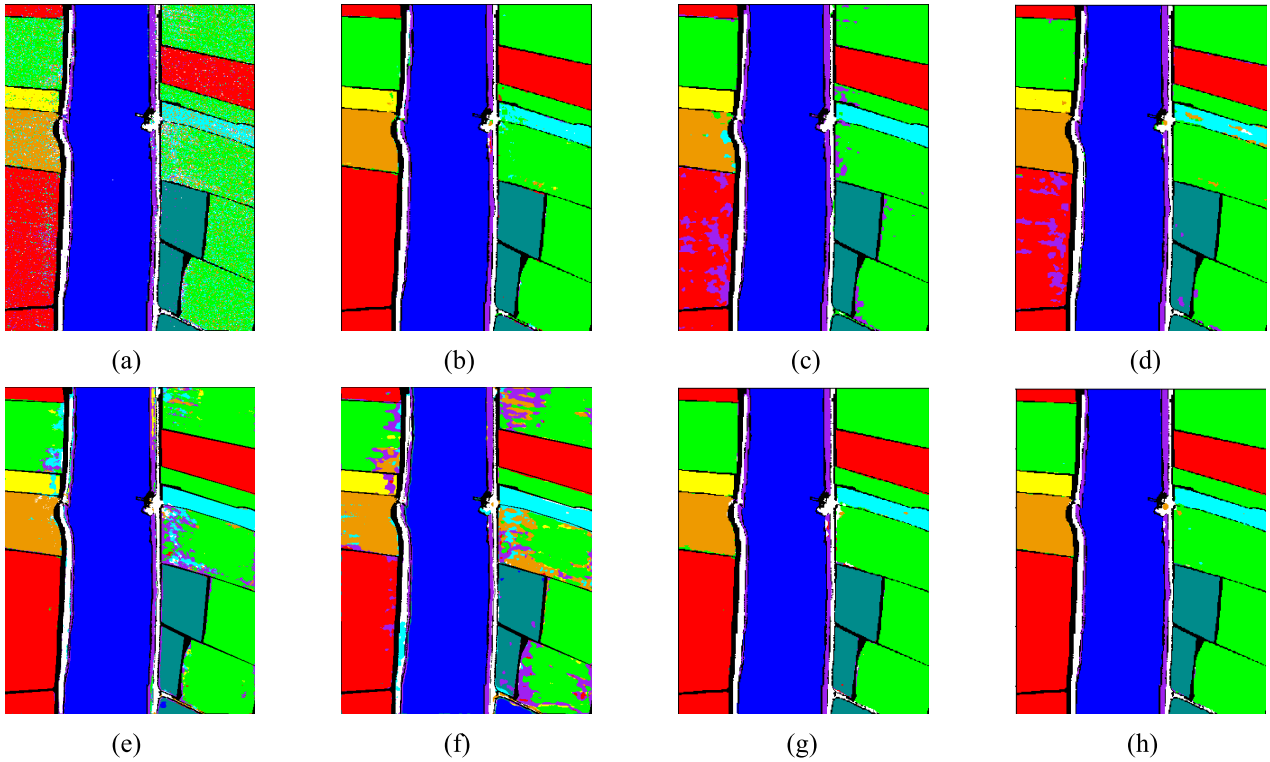


Fig. 6. Classification maps obtained via different algorithms for the WHU-Hi-LongKou dataset. (a) EGNN. (b) FADCNN. (c) SSRN. (d) FDSSC. (e) DCFSL. (f) CMFSL. (g) PN. (h) Ours.

TABLE II
COMPARISON OF THE PER-CLASS ACCURACY, OA, AA, AND KAPPA OF DIFFERENT ALGORITHMS FOR THE WHU-HI-LONGKOU DATASET

Class	EGNN	FADCNN	SSRN	FDSSC	DCFSL	CMFSL	PN	Ours
C1	90.11 ± 3.28	99.80 ± 0.01	98.72 ± 0.06	99.83 ± 0.00	99.54 ± 0.14	98.27 ± 0.28	99.62 ± 0.20	**99.85 ± 0.08**[1]
C2	84.42 ± 1.36	98.48 ± 0.04	78.72 ± 0.00	87.31 ± 0.12	96.26 ± 0.75	86.38 ± 7.32	98.43 ± 0.66	**99.49 ± 0.15**
C3	89.34 ± 3.87	96.28 ± 0.17	85.68 ± 0.14	99.74 ± 0.00	90.22 ± 0.20	84.84 ± 1.76	99.27 ± 0.20	**100.00 ± 0.00**
C4	64.72 ± 11.99	99.12 ± 0.02	92.62 ± 0.03	99.03 ± 0.01	87.71 ± 0.18	72.13 ± 0.37	98.74 ± 0.67	**99.30 ± 0.24**
C5	77.19 ± 0.51	88.26 ± 0.22	77.45 ± 0.01	88.32 ± 0.09	97.35 ± 0.70	94.33 ± 1.78	97.03 ±1.01	**97.81 ± 0.34**
C6	99.00 ± 3.27	98.72 ± 0.02	98.32 ± 0.00	99.08 ± 0.01	92.59 ± 1.77	87.63 ± 0.76	98.41 ± 0.17	**99.55 ± 0.23**
C7	99.87 ± 1.06	**99.97 ± 0.00**	**99.97 ± 0.00**	99.85 ± 0.00	99.76 ± 0.06	98.53 ± 0.24	99.82 ± 0.08	99.95 ± 0.03
C8	94.40 ± 1.74	**93.33 ± 0.11**	87.49 ± 0.13	97.54 ± 0.02	88.77 ± 3.83	85.28 ± 3.51	89.31 ± 2.88	92.06 ± 0.79
C9	92.12 ± 2.88	94.28 ± 0.10	60.66 ± 0.06	68.50 ± 0.22	82.24 ± 5.80	84.22 ± 1.08	97.12 ± 1.55	**99.13 ± 0.16**
OA	85.67 ± 2.88	98.88 ± 0.01	90.68 ± 0.09	97.00 ± 0.02	94.44 ± 0.05	88.08 ± 0.31	98.81 ± 0.37	**99.35 ± 0.11**
AA	87.91 ± 1.64	96.47 ± 0.04	86.63 ± 0.10	93.24 ± 0.04	92.77 ± 0.10	87.95 ± 0.74	97.53 ± 0.56	**98.57 ± 0.12**
Kappa	81.93 ± 3.49	98.52 ± 0.01	88.06 ± 0.11	96.08 ± 0.02	92.78 ± 0.05	88.14 ± 4.82	98.45 ± 0.47	**99.14 ± 0.14**

selected classes N_c, the number of support samples N_s, and the number of query samples N_q for each episode are all set to 5, which is devised to mimic the few-shot task and make the training problem more in accord with the test environment.

In the experimental setting, the learning rate is fixed to 0.001, the number of training iterations is 3000, and the Adam optimizer is adopted to guide parameter updating optimization. In especial, the number of supervised samples L is set to 5, the regularization coefficients γ and η are set to 0.5, and 1, respectively, and the margin in the contrastive loss α is set to 0.2. In the CD of the supervised samples process, the hyperparameter ε is, finally, set to 0.01, and promising results are obtained.

To evaluate the performance of the proposed model, five state-of-the-art methods for few-shot HSIC are exploited for comparison, including EGNN [48], FADCNN [49], SSRN [50], FDSSC [51], DCFSL [33], CMFSL [52], and PN [30]. In particular, the hyperparameters setting of the compared methods is the same as the original papers.

C. Result and Analysis

In this section, we display the experimental results of the proposed MCPN and the comparative methods on the three popular datasets. The classification performance with overall accuracy (OA), average accuracy (AA), and Kappa is given in Tables II–IV. The colorful result maps are demonstrated in Figs. 6–8.

The first experiment is conducted on the WHU-Hi-LongKou dataset. As shown in Table II, MCPN achieves the highest OA of 99.35%. It can be analyzed that the multiview rectifying strategies have a positive effect on the prototypical network. The colorful result maps in Fig. 6 show the results in a

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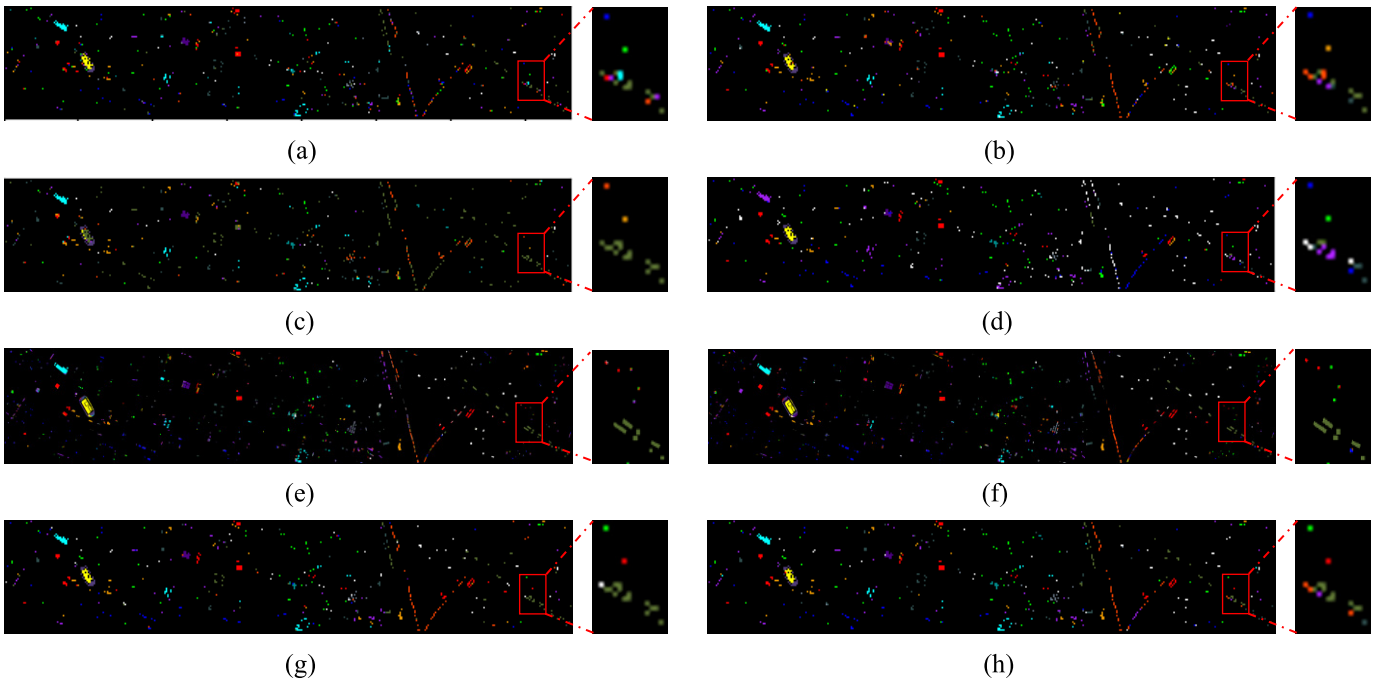


Fig. 7. Classification maps obtained via different algorithms for the Houston dataset. (a) EGNN. (b) FADCNN. (c) SSRN. (d) FDSSC. (e) DCFSL. (f) CMFSL. (g) PN. (h) Ours.

TABLE III

COMPARISON OF THE PER-CLASS ACCURACY, OA, AA, AND KAPPA OF DIFFERENT ALGORITHMS FOR THE HOUSTON DATASET

Class	EGNN	FADCNN	SSRN	FDSSC	DCFSL	CMFSL	PN	Ours
C1	79.70 ± 4.44	76.74 ± 0.42	91.19 ± 0.22	89.19 ± 0.08	88.64 ± 0.36	84.75 ± 0.58	88.39 ± 4.88	**94.74 ± 1.53**
C2	60.29 ± 4.22	82.54 ± 0.59	60.96 ± 0.41	71.99 ± 0.17	60.97 ± 0.12	65.46 ± 1.80	84.26 ± 7.29	**89.31 ± 5.28**
C3	89.96 ± 4.67	92.80 ± 0.55	96.84 ± 0.08	90.88 ± 0.13	90.25 ± 1.38	92.58 ± 1.00	99.23 ± 0.22	**99.58 ± 0.34**
C4	66.88 ± 14.31	92.56 ± 0.53	52.54 ± 0.31	85.45 ± 0.14	74.94 ± 1.66	53.89 ± 1.32	92.88 ± 1.66	**94.86 ± 1.82**
C5	92.19 ± 5.91	93.4 ± 0.44	69.84 ± 0.39	96.77 ± 0.04	87.43 ± 1.50	81.08 ± 1.38	97.83 ± 1.17	**99.16 ± 0.57**
C6	54.77 ± 3.20	31.32 ± 1.13	88.48 ± 0.37	**93.63 ± 0.07**	60.16 ± 4.85	44.89 ± 2.73	89.23 ± 3.47	91.69 ± 2.07
C7	42.03 ± 7.27	58.97 ± 0.60	93.07 ± 0.33	72.10 ± 0.17	55.98 ± 3.57	59.33 ± 1.43	82.51 ± 2.19	**85.09 ± 6.32**
C8	48.55 ± 6.72	66.33 ± 0.69	63.44 ± 0.34	66.83 ± 0.21	47.54 ± 4.67	43.61 ± 1.12	39.83 ± 7.67	**77.04 ± 13.39**
C9	45.77 ± 7.33	69.63 ± 0.75	58.58 ± 0.15	70.16 ± 0.18	60.83 ± 0.77	50.90 ± 2.78	74.31 ± 1.93	**84.56 ± 1.32**
C10	77.75 ± 7.30	55.35 ± 0.48	46.24 ± 0.33	**92.16 ± 0.08**	63.47 ± 0.21	55.86 ± 0.90	81.40 ± 9.26	76.69 ± 8.48
C11	43.48 ± 2.55	72.15 ± 0.29	73.00 ± 0.28	**93.28 ± 0.06**	62.03 ± 2.36	69.59 ± 1.40	75.11 ± 1.42	82.78 ± 4.45
C12	85.97 ± 7.87	74.47 ± 0.68	55.42 ± 0.39	74.87 ± 0.25	44.02 ± 2.07	62.87 ± 5.79	78.67 ± 4.71	**85.16 ± 4.41**
C13	56.08 ± 6.84	22.39 ± 1.12	80.42 ± 0.32	70.48 ± 0.37	70.58 ± 4.85	77.59 ± 1.61	80.68 ± 5.05	**85.82 ± 5.20**
C14	80.84 ± 1.77	68.44 ± 1.70	97.61 ± 0.40	98.64 ± 0.03	83.33 ± 2.01	96.06 ± 0.80	**97.20 ± 1.69**	97.08 ± 1.26
C15	63.33 ± 4.40	89.53 ± 1.59	82.39 ± 0.02	92.04 ± 0.08	79.09 ± 4.89	48.60 ± 3.83	91.88 ± 2.05	**98.22 ± 0.66**
OA	65.36 ± 3.07	73.03 ± 0.18	53.57 ± 0.27	75.68 ± 0.11	67.04 ± 0.31	64.67 ± 0.47	81.72 ± 1.59	**87.56 ± 2.10**
AA	65.83 ± 2.92	69.76 ± 0.16	73.84 ± 0.21	83.90 ± 0.07	68.62 ± 0.49	65.80 ± 0.32	83.55 ± 1.12	**87.78 ± 1.94**
Kappa	62.54 ± 2.19	70.81 ± 0.20	49.71 ± 0.29	73.66 ± 0.12	64.38 ± 0.33	61.70 ± 0.30	80.25 ± 1.71	**86.55 ± 2.27**

more intuitive pattern. In particular, the classification accuracy of classes 1–4, 6, 7, and 9 is more than 99%, while the accuracy of classes 5 and 8 is slightly lower. Compared with Figs. 3(b) and 6(h), it can be observed that the wrong classification is mainly caused by the boundary pixels of the narrow-leaf soybeans as the broad-leaf soybeans and the misclassification of some pixels of class 8 as class 2.

The second classification results on the Houston dataset are demonstrated in Table III, and the colorful result maps are illustrated in Fig. 7. It is obvious that the proposed framework is superior to other methods. From Table III, we observe that MCPN (87.56%) performs better than PN (81.72%), proving that the calibrated modules are essential and beneficial to the proposed architecture. In addition, in terms of the accuracy of each category, although the classification accuracy of class 8 is less than 80%, it has greatly improved compared with other algorithms. Combined with Table III and Fig. 7, we can conclude that the main reason for the OA of 76.69% for class 10 is that some pixels are mistakenly divided into class 12.

Finally, Table IV shows the classification accuracy of Botswana acquired through different methods. Fig. 8 displays the colorful classification maps of these models. Table IV illustrates that our framework is beneficial for the accurate classification of Botswana. The OA of MCPN was 5.34%, 3.36%, 2.26%, 1.95%, 5.59%, 8.37%, and 1.86% higher than that of EGNN, FADCNN, SSRN, FDSSC, DCFSL, CMFSL, and PN, respectively. From the experimental results in Table IV, MCPN obtains better performance than other

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TABLE IV
COMPARISON OF THE PER-CLASS ACCURACY, OA, AA, AND KAPPA OF DIFFERENT ALGORITHMS FOR THE BOTSWANA DATASET

Class	EGNN	FADCNN	SSRN	FDSSC	DCFSL	CMFSL	PN	Ours
C1	**100.00± 0.00**	**100.00 ± 0.00**	99.63 ± 0.00	99.85 ± 0.00	**100.00 ± 0.00**	99.00 ± 0.36	**100.00 ± 0.00**	**100 .00± 0.00**
C2	98.02 ± 0.69	91.88 ± 1.92	88.74 ± 0.23	**100.00 ± 0.00**	97.92 ± 1.47	99.65 ± 0.49	99.80 ± 0.40	**100.00 ± 0.00**
C3	95.22 ± 0.28	94.74 ± 0.77	97.43 ± 0.05	**100.00 ± 0.02**	94.58 ± 1.53	73.58 ± 0.33	99.12 ± 1.08	99.60 ± 0.69
C4	91.63 ± 3.61	95.63 ± 0.47	98.29 ± 0.02	98.86 ± 0.00	96.98 ± 1.80	89.21 ± 1.25	99.72 ± 0.56	**100.00 ± 0.00**
C5	89.22 ± 3.95	90.63 ± 0.89	98.77 ± 0.02	**99.78** ± 0.02	87.63 ± 3.57	94.44 ± 1.25	94.50 ± 3.16	98.70 ± 0.42
C6	89.96 ± 1.32	94.87 ± 0.49	**99.77** ± 0.00	98.73 ± 0.00	67.68 ± 2.68	51.77 ± 1.39	94.87 ± 4.06	98.98 ± 0.16
C7	94.21 ± 1.91	98.61 ± 0.52	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.0 ± 00.00**	**100.00 ± 0.00**	98.46 ± 1.48	99.81 ± 0.33
C8	99.01 ± 0.70	93.89 ± 0.39	99.80 ± 0.00	**100.00 ± 0.11**	91.41 ± 12.14	**100.00 ± 0.00**	98.72 ± 2.32	99.75 ± 0.43
C9	89.81 ± 1.12	92.23 ± 0.38	98.10 ± 0.04	93.40 ± 0.10	95.68 ± 2.14	98.76 ± 0.59	95.35 ± 1.35	**99.92 ± 0.14**
C10	91.53 ± 0.86	**100.00 ± 0.00**	97.35 ± 0.05	88.02 ± 0.10	98.08 ± 1.36	92.87 ± 2.57	99.60 ± 0.81	99.90 ± 0.17
C11	95.74 ± 3.71	**99.80 ± 0.16**	96.45 ± 0.06	99.74 ± 0.01	99.33 ± 0.47	95.89 ± 1.66	98.56 ± 2.08	99.76 ± 0.42
C12	99.45 ± 1.95	97.13 ± 0.81	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	98.46 ± 1.92	99.86 ± 0.24
C13	97.01 ± 0.26	**100.00 ± 0.00**	95.86 ± 0.08	**100.00 ± 0.00**	**100.00 ± 0.00**	95.82 ± 2.42	97.16 ± 2.38	99.91 ± 0.16
C14	96.84 ± 2.23	96.84 ± 0.66	**100.00 ± 0.00**	**100.00 ± 0.00**	88.15 ± 8.38	87.78 ± 0.00	98.32 ± 2.87	**100.00 ± 0.00**
OA	94.36 ± 0.68	96.34 ± 0.20	97.44 ± 0.04	97.75 ± 0.02	94.11 ± 0.87	91.33 ± 0.47	97.84 ± 1.22	**99.70 ± 0.16**
AA	94.83 ± 0.77	96.16 ± 0.26	97.87 ± 0.03	98.46 ± 0.01	94.10 ± 1.01	91.33 ± 0.37	98.04 ± 1.25	**99.73 ± 0.15**
Kappa	93.90 ± 0.73	96.03 ± 0.21	97.24 ± 0.04	97.56 ± 0.02	93.61 ± 0.95	90.17 ± 0.45	97.66 ± 1.32	**99.67 ± 0.17**

TABLE V
ABLATION STUDY ON THE THREE HSI DATASETS

Module Description	Baseline	√	√	√	√	√	√	√	√	√	√	√
	CPR		√				√	√	√			√
	CD			√			√			√	√	√
	J_{DL}				√			√		√		√
	J_C					√			√		√	√
HSI DataSet	WHU-Hi-LongKou	98.81	98.95	99.07	99.29	99.06	99.10	99.31	99.07	99.32	99.20	99.35
	Houston	81.72	82.51	84.98	83.81	80.78	85.26	85.49	81.83	85.87	84.86	87.56
	Botswana	97.84	98.52	98.63	99.26	98.65	99.29	99.38	98.87	99.54	98.90	99.70

algorithms due to significantly improving the classification accuracy of class 9. Particularly, the AA, OA, and Kappa of MCPN get 1.86%, 1.69%, and 2.01% improvements over PN, respectively, which indicates that the feature enhancement ability of the multiview calibrated modules plays a positive role for Botswana in HSIC.

D. Ablation Study

To demonstrate the contribution of the CPR, CD, J_{DL}, and J_C modules in the proposed MCPN framework to the final classification results, we perform ablation studies on three datasets with a baseline network of PN. The experimental results are listed in Table V.

As demonstrated in Table V, the OA of the three datasets is improved to varying degrees with CPR, CD, and J_{DL} adding to the baseline, respectively. In particular, the classification results of the module CPR + CD are better than that of a single module for all three datasets. The OA of CPR + CD for WHU-Hi-LongKou, Houston, and Botswana datasets is increased by 0.15% and 0.03%, 2.75% and 0.28%, and 0.77% and 0.66%, respectively. It shows that both modules are effective, and the combination achieves better results. In addition, the module CPR + J_{DL} and the module CD + J_{DL} both produce excellent performances on three datasets ulteriorly demonstrating the effectiveness of the three components in our framework. However, the OA of the J_C module is 0.94% fewer than that of the baseline for the Houston dataset, and the experimental results of CPR + J_C, CD + J_C are lower than applying the CPR or CD module alone, indicating that the J_C module is not competitive to the HSIC of the Houston dataset. Furthermore, additional studies show that the module CPR + CD + J_{DL} + J_C is the most competitive, especially for the Houston dataset, which increases by 5.84% over the baseline, and the OA of WHU-Hi-LongKou and Botswana datasets is improved by 0.54% and 1.86%, respectively. With the analysis above, we can conclude that the multiview rectification pattern proposed in this article is an effective solution, whether for datasets with sparse or dense sample distribution.

E. Analysis of Experimental Parameters

In this article, we carry out experiments and analyze the effects of various parameters on classification accuracy, as illustrated in Table VI and Figs. 9–12, including the number of supervised samples L, the regularization coefficient of discriminate loss γ , the regularization coefficient of contrastive loss η, the margin in the contrastive loss α, and the hyperparameter ε in the CD.

The number of supervised samples L is set to {1, 3, 5, 10, 15, 20, 25} during the testing process. The experimental results for the three datasets are shown in Table VI. Despite the OA varying within a small range for WHU-Hi-LongKou, Houston, and Botswana datasets, we can still observe that the OA rises as L increases, which means that more supervised samples provide more information, and better results

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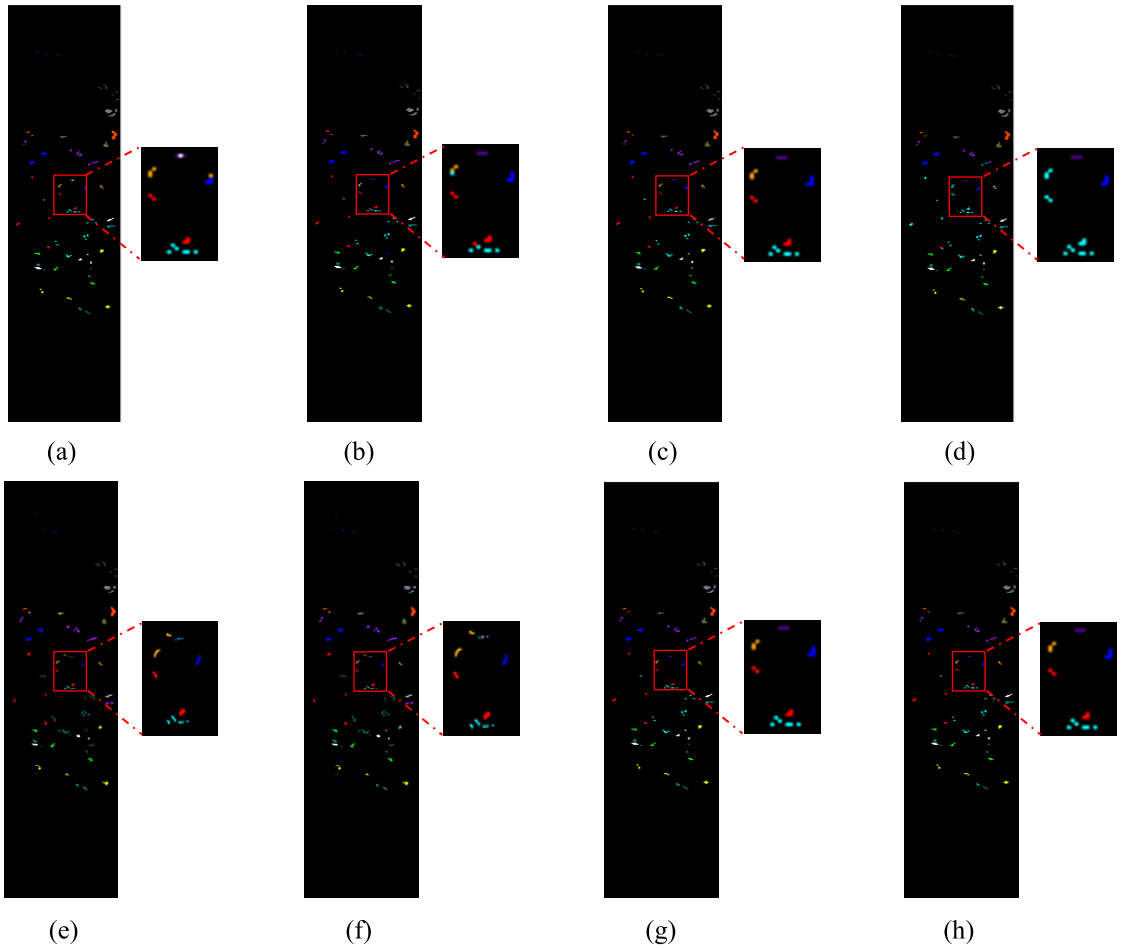


Fig. 8. Classification maps obtained via different algorithms for the Botswana dataset. (a) EGNN. (b) FADCNN. (c) SSRN. (d) FDSSC. (e) DCFSL. (f) CMFSL. (g) PN. (h) Ours.

TABLE VI
OA (%) WITH DIFFERENT NUMBERS OF SUPERVISED SAMPLES FOR THE EXPERIMENTAL DATASETS

Dataset \ L	1	3	5	10	15	20	25
WHU-Hi-LongKou	99.28	99.30	99.35	99.37	99.42	99.40	99.43
Houston	83.39	85.03	87.56	87.66	88.24	88.81	88.69
Botswana	99.27	99.42	99.70	99.76	99.77		

are achieved. Notably, the experiments are performed on the Botswana dataset with L in the setting of {1, 3, 5, 10, 15} since some categories have less than 20 training samples. As can be observed, with the increment of the number of supervised samples, the OA of the three datasets generally shows an upward trend. Especially, the OA with the setting of L of 25 is slightly higher than the setting of L of 5 for WHU_Hi_LongKou, while the OA with L of 15 yields the best performance in the Houston dataset. The highest accuracy is acquired with L of 20 for Botswana. In this article, L is set to 5 to better simulate the few-shot task in other experiments.

As shown in Fig. 9, the regularization coefficient of discriminate loss γ is set from 0.1 to 1, and one experiment is performed on each dataset. The column chart shows that the highest and lowest accuracies differ by 3.86%, 0.78%, and 0.31% for Houston, WHU-Hi-LongKou, and Botswana datasets, respectively. As demonstrated in the chart, when γ equals 0.5, relatively high classification results are generated for all three datasets. Therefore, γ is set to 0.5 in other experiments of this article.

Next, we analyze the influence of the regularized term η in the contrastive loss on the performances of the proposed model. Specifically, we set the variation of the regularization coefficient from 0.1 to 1 with an interval of 0.1, and the accuracies results of the three datasets with different η's are demonstrated in Fig. 10. As can be observed, the OA of the Houston and Botswana datasets reaches the maximum when η is set to 1, which is 3.26% and 0.62% higher than the minimum values, respectively, while, for the WHU-Hi-LongKou dataset, the accuracy is highest when η equals to 0.8. It is 0.12% higher than the accuracy at η of 1. For the competitive comparison, the value of η is set to 1 in the above experiments for all the datasets.

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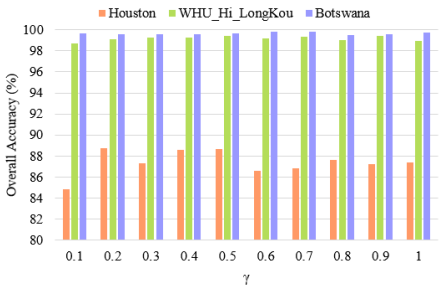


Fig. 9. OA with different coefficients of the discriminant loss for WHU-Hi-LongKou, Houston, and Botswana datasets.

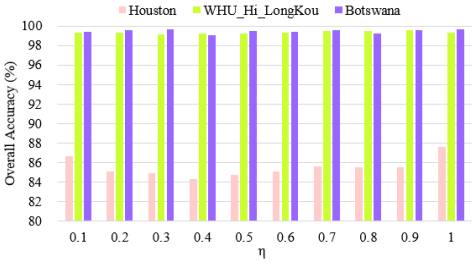


Fig. 10. OA with different coefficients of the contrastive loss for WHU-Hi-LongKou, Houston, and Botswana datasets.

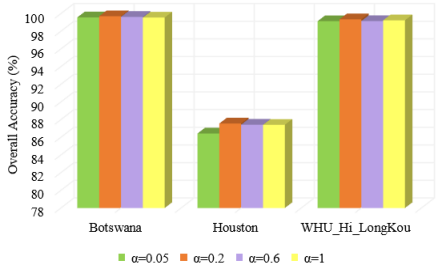


Fig. 11. OA with different margins in the contrastive loss for WHU-Hi-LongKou, Houston, and Botswana datasets.

Besides, we conduct experiments to discover the effect of different α's in the metric learning module. The margin α in the contrastive loss is set to {0.05, 0.2, 0.6, 1} for analysis. From the experimental results in Fig. 11, we observe that, when α is set to 0.2, the overall classification accuracy of the three datasets reaches the maximum value, which is 1.13%, 0.21%, and 0.15% higher than the minimum accuracy, respectively. It suggests that too large or too small margins have a restrictive effect on the final performance.

Moreover, in the CD of supervised samples, we analyze the effect of a different hyperparameter ε on classification accuracy with ε in the set of {0.01, 0.1, 0.3}. As illustrated in Fig. 12, when ε is equal to 0.01, the OA achieves the highest. In other words, the increasing ε has an inhibitory effect on the experimental results consequently, and a smaller dispersion of features is more effective in calibrating the feature distribution.

F. Time Consumption

The training time comparison of the proposed method and other popular algorithms on the three datasets is shown in Fig. 13. The computer configuration for training time calculation of different methods in this article is the same. As shown in Fig. 13, FDSSC has the shortest training time due to the

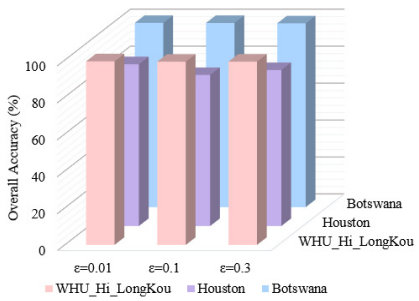


Fig. 12. OA with different hyperparameters in CD for WHU-Hi-LongKou, Houston, and Botswana datasets.

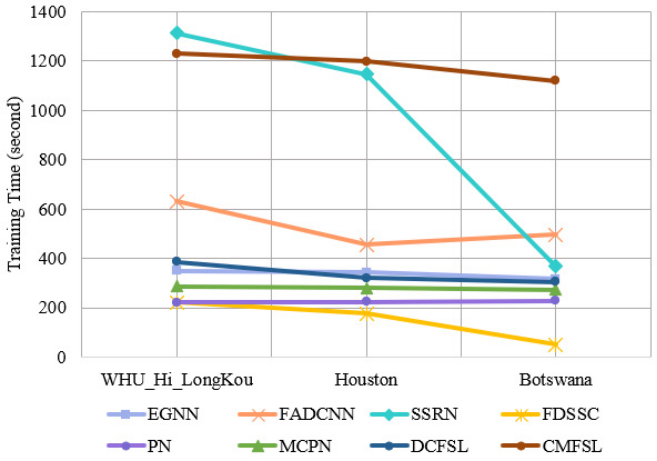


Fig. 13. Time consumption of the training process for WHU-Hi-LongKou, Houston, and Botswana datasets.

characteristics of the model design and the advantages of fewer epochs. Compared with PN, the MCPN method generates better performance with three more calibrated modules, which inevitably costs more calculation and training times than the original PN model. The joining of several calibrated modules is the factor that causes the training time of MCPN to be longer than that of PN. The other comparison methods have a long training time, which suggests that the method proposed in this article has an advantage in both training time and classification accuracy.

IV. Conclusion

Prototype representation for the few-shot HSIC obtained more attention due to the representative knowledge of the specific category. In this study, a new MCPN model is proposed for few-shot HSIC based on multiview calibrated prototype learning. The main advantage of the presented model is to acquire calibrated prototypes by making full use of feature information from multiple angles that consist of local aggregation adjustment, metric learning regularization, and feature distribution calibration, in which the representation of the original prototype is modified with local HSI patch aggregation through the presented CAN network. The metric learning regularization is accomplished with three designed crucial losses. The feature distribution is calibrated with prior statistical knowledge in the test phase. Sufficient experiment results and analysis demonstrate that our proposed framework achieves excellent performance on public HSI datasets.

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The calibrated prototype learning approach for HSIC infers unsatisfactory performance by the approximative spectrum, which is caused by the complexity of the spatial–spectral expression with the limited samples. To pursue better performance for the few-shot HSIC in the severe scene, we will explore effective strategies to improve the sample diversity and fine-grained prototype representation in future work.

REFERENCES

[1] P. Niu and W. Wei, "Classification of hyperspectral remote sensing images with dynamic support vector machine ensemble," *J. Comput. Appl.*, vol. 30, no. 6, pp. 1590–1593, Jun. 2010.

[2] J. Jiang, L. Huang, H. Li, and L. Xiao, "Hyperspectral image supervised classification via multi-view nuclear norm based 2D PCA feature extraction and kernel ELM," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1496–1499.

[3] W. Di and M. M. Crawford, "Active learning via multi-view and local proximity co-regularization for hyperspectral image classification," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 618–628, Jun. 2011.

[4] B. Baassou, H. Mingyi, M. I. Farid, and M. Shaohui, "Hyperspectral image classification based on iterative support vector machine by integrating spatial–spectral information," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 1023–1026.

[5] G. Licciardi, P. R. Marpu, J. Chanussot, and J. A. Benediktsson, "Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 447–451, May 2012.

[6] C. Yu et al., "Hyperspectral image classification method based on CNN architecture embedding with hashing semantic feature," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1866–1881, Jun. 2019.

[7] X. He and Y. Chen, "Optimized input for CNN-based hyperspectral image classification using spatial transformer network," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1884–1888, Dec. 2019.

[8] J. Feng et al., "Attention multibranch convolutional neural network for hyperspectral image classification based on adaptive region search," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5054–5070, Jun. 2021.

[9] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[10] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[11] X. Sun et al., "Ensemble-based information retrieval with mass estimation for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–23, 2022.

[12] W. Rao et al., "Transferable network with Siamese architecture for anomaly detection in hyperspectral images," *Int. J. Appl. Earth Observat. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102669.

[13] W. Hu, Y. Huang, W. Li, F. Zhang, and H.-C. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[14] P. Jia, M. Zhang, W. Yu, F. Shen, and Y. Shen, "Convolutional neural network based classification for hyperspectral data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5075–5078.

[15] M. Paoletti et al., "Deep pyramidal residual networks for spectral-spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.

[16] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral-spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.

[17] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[18] J.-Y. Yang, H.-C. Li, W.-S. Hu, L. Pan, and Q. Du, "Adaptive cross-attention-driven spatial–spectral graph convolutional network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[19] W. Rao et al., "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022.

[20] X. Sun et al., "Target detection through tree-structured encoding for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4233–4249, May 2021.

[21] X. Zhang, P. Weng, J. Feng, E. Zhang, and B. Hou, "Spatial-spectral classification based on group sparse coding for hyperspectral image," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 1745–1748.

[22] X. Tang et al., "Hyperspectral image classification based on 3-D octave convolution with spatial–spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

[23] M. Kanthi, T. H. Sarma, and C. S. Bindu, "A 3D-deep CNN based feature extraction and hyperspectral image classification," in *Proc. IEEE India Geosci. Remote Sens. Symp. (InGARSS)*, Dec. 2020, pp. 229–232.

[24] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. ICML Deep Learn. Workshop*, Mar. 2017, pp. 1126–1135.

[25] M. A. Jamal and G.-J. Qi, "Task agnostic meta-learning for few-shot learning," in *Proc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11711–11719.

[26] S. Ravi and H. Larochelle, "Optimization as a model for few-shot learning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Nov. 2017, pp. 1–11.

[27] T. Elsken, B. Staffler, J. H. Metzen, and F. Hutter, "Meta-learning of neural architectures for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12362–12372.

[28] G. Koch, R. Zemel, and R. Salakhutdinov, "Siamese neural networks for one-shot image recognition," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 2, 2015, pp. 1–30.

[29] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Neural Inf. Process. Syst. (NIPS)*, Jun. 2016, pp. 1–9.

[30] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Neural Inf. Process. Syst. (NIPS)*, Mar. 2017, pp. 1–11.

[31] Y. Li, L. Zhang, W. Wei, and Y. Zhang, "Deep self-supervised learning for few-shot hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 501–504.

[32] F. Zhou, L. Zhang, W. Wei, Z. Bai, and Y. Zhang, "Meta transfer learning for few-shot hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 3681–3684.

[33] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[34] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5085–5102, Jun. 2021.

[35] K. Huang, X. Deng, J. Geng, and W. Jiang, "Self-attention and mutual-attention for few-shot hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2230–2233.

[36] X. Ma, S. Ji, J. Wang, X. Liu, and H. Wang, "Classification of hyperspectral image based on task-specific learning network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8646–8656, Oct. 2021.

[37] X. Tong, J. Yin, B. Han, and H. Qv, "Few-shot learning with attention-weighted graph convolutional networks for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 1686–1690.

[38] W. Wang, Y. Chen, X. He, and Z. Li, "Soft augmentation-based Siamese CNN for hyperspectral image classification with limited training samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[39] L. Huang and Y. Chen, "Dual-path Siamese CNN for hyperspectral image classification with limited training samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 518–522, Mar. 2021.

[40] B. Xi, J. Li, Y. Li, and Q. Du, "Semi-supervised graph prototypical networks for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2851–2854.

[41] X. Li, Z. Cao, L. Zhao, and J. Jiang, "ALPN: Active-learning-based prototypical network for few-shot hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[42] S. Wang, B. Du, D. Zhang, and F. Wan, "Adversarial prototype learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[43] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:30:06 UTC from IEEE Xplore. Restrictions apply.

[44] C. Zhang, J. Yue, and Q. Qin, "Global prototypical network for few-shot hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4748–4759, 2020.
[45] H. Tang, Y. Li, X. Han, Q. Huang, and W. Xie, "A spatial–spectral prototypical network for hyperspectral remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 167–171, Jan. 2020.
[46] B. Xi, J. Li, Y. Li, R. Song, and Q. Du, "Deep prototypical networks with hybrid residual attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3683–3700, 2020.
[47] S. Yang, L. Liu, and M. Xu, "Free lunch for few-shot learning: Distribution calibration," 2021, *arXiv:2101.06395*.
[48] J. Kim, T. Kim, S. Kim, and C. D. Yoo, "Edge-labeling graph neural network for few-shot learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 11–20.
[49] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "Feedback attention-based dense CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.
[50] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral-spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
[51] W. Wang, D. Shuguang, J. Zhongmin, and S. Liujie, "A fast dense spectral-spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, Jul. 2018.
[52] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.



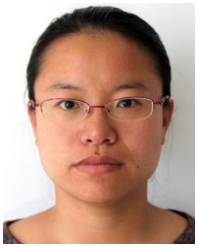
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